Closed-form Second-order Data Shapley for Group Fairness

Down-weighting biased training data, without group labels

A model inherits the bias in its data. We use a closed-form data-value score to stop it — **during training**, with no group labels in the update.

Shown across three kinds of data bias: color, texture, natural images.

Bias in the data becomes unfairness in the model

- In training, a **majority group** ties an attribute to the label — background, color, or a demographic feature.
- An ordinary model uses that **biased attribute**. Average accuracy looks fine, but it **fails the minority group** where the attribute does not hold.
- That is an **unfair model**: good for the well-represented, poor for the under-represented.

> Strong fixes need extra help: **group labels** (GroupDRO), a **second stage** (JTT), or a **bi-level loop** (meta-reweighting).

Our question

Can a **closed-form, single-pass** signal — computed during normal training, with **no group labels in the update** — find the examples that teach the bias, so we can down-weight them?

Idea: value each example, then down-weight the biased ones

- **In-Run Data Shapley** gives each training example a **closed-form score**: how much it helps a small, **group-balanced** reference set — computed *during* training.
- An example that helps the biased majority but **not** the balanced set gets a **low score**.
- We turn the score into a weight and do weighted training. Low score ⇒ less weight.

$$w_i = B \cdot \mathrm{softmax}(\ \mathrm{EMA}(\varphi_i) / \tau\ ) \ \rightarrow\ \theta_{t+1} = \theta_t - \eta \sum_i w_i \nabla L(z_i)$$

The question becomes: **what score φ_i best targets the bias?**

Method — closed-form Shapley score (1st & 2nd order)

$$\varphi_i^{(1)} = \langle\, g_i, g_{val} \,\rangle \quad \rightarrow \quad \varphi_i^{(2)} = \varphi_i^{(1)} - \alpha \cdot \langle\, g_i, H_{val}\, g_{val} \,\rangle$$

- **1st order:** $\varphi^{(1)}$ = how much z_i helps the balanced set in one step. Per-sample gradients via **torch.func.vmap** — closed form, no bi-level loop.
- **2nd order:** add the gradient–Hessian–gradient cross-term via the **Pearlmutter trick** (one extra backward pass).
- **Key trick:** rescale the cross-term to the **same RMS magnitude** as $\varphi^{(1)}$ each minibatch.

⚠ **Without RMS rescaling, the method collapses under strong bias.**

Does the 2nd-order term really target bias?

A fair worry: maybe the curvature term is just **smoothing**, not a real bias signal. We checked.

The geometry

- g_{val} is almost the **top eigenvector of H_{val}** ⇒ the cross-term is **nearly parallel** to $\varphi^{(1)}$.
- Split it: cross = $\boldsymbol{\beta \cdot \varphi^{(1)}}$ (parallel = global shrinkage) + **r** (orthogonal residual).

> The parallel part is just smoothing. **Is the residual r a real bias signal — or noise?**

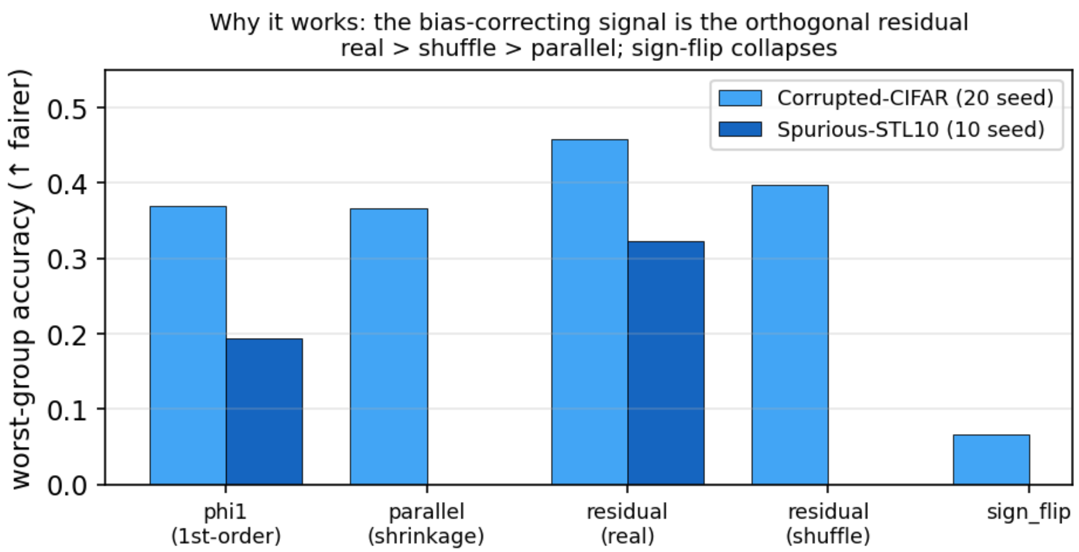
The bias-correcting signal is the residual

Five arms share $\varphi^{(1)}$, treat the residual *r* differently:

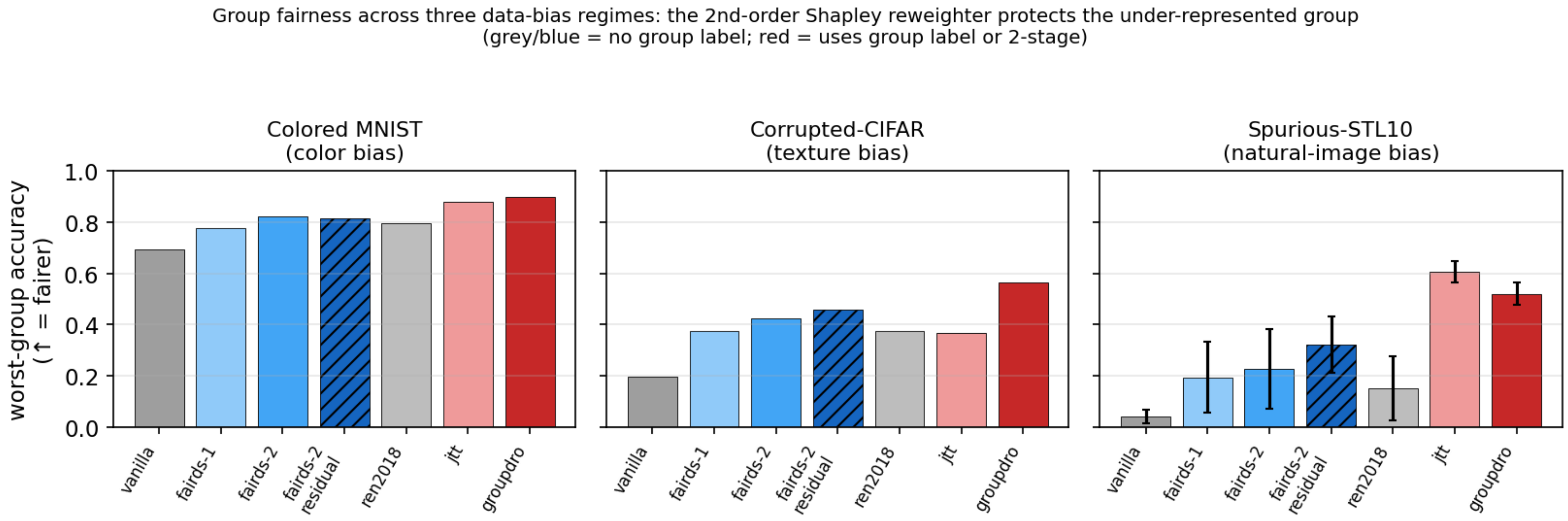
- **residual (real):** $\varphi^{(1)} - \alpha \cdot r$
- **residual (shuffle):** r permuted within bins
- **parallel:** shrinkage only
- **sign_flip:** $\varphi^{(1)} + \alpha \cdot r$

real > shuffle > parallel;
sign-flip collapses.

CIFAR: real vs shuffle **+5.9pp, p=0.0015**. If *r* were noise, shuffling would not matter — it does.

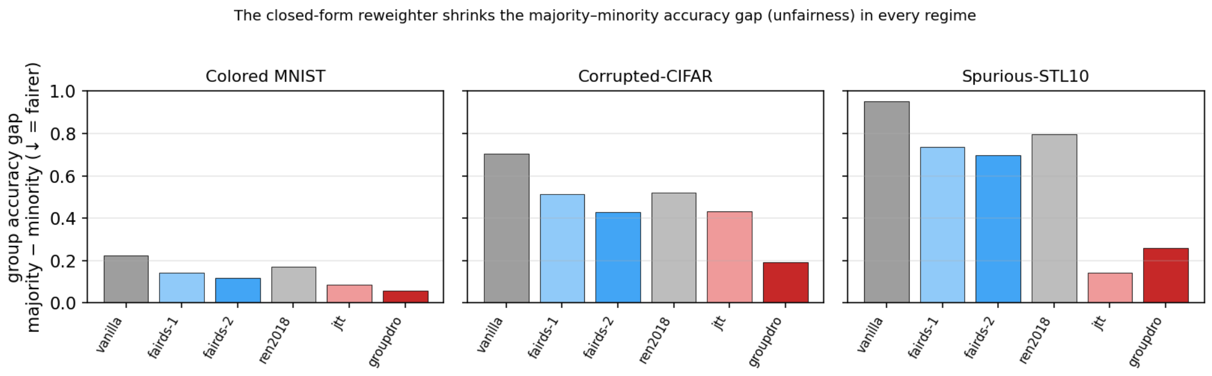


Group fairness across three data-bias regimes



Worst-group accuracy (protecting the minority). Our reweighter (striped) beats every **no-group-label** baseline in **all three** bias types — color, texture, natural images. Group-label / 2-stage methods (red) are fairer still.

It shrinks the unfairness gap — by lifting the minority



- Majority−minority accuracy gap **cut 40–47%** vs ERM (CMNIST .223→.118, CIFAR .703→.430, STL .950→.568).
- The gap closes by **helping the minority**, not hurting the majority: STL minority **0.04**→**0.32**, majority only 0.99→0.89, overall ↑ 0.14→0.38.
- **Honest:** JTT / GroupDRO are fairer, but use a 2nd stage or group labels.

> This is the behavior fairness asks for: protect the under-represented group.

Demographic fairness (Adult, COMPAS) — honest scope

data	method	DP ↓	EO ↓	acc
Adult	vanilla	0.196	0.117	0.837
	Fairds-2	0.189	0.127	0.835
	Ren2018 (bi-level)	**0.175**	**0.102**	0.819

- On tabular demographic data, Fairds **keeps accuracy** but barely moves DP/EO (not significant).
- The bi-level reweighter is fairer, at an accuracy cost.
- **Why:** tabular bias is **diffuse** (many weak features) ⇒ the per-example signal is small.

> **Scope:** the mechanism needs **structured** bias (one dominant attribute) to fire — strong on the image regimes, weak on diffuse tabular bias.

Cost & practicality

Wall-clock overhead (ResNet-18)

method	×vanilla
Fairds-1	4.30×
Ren2018 (bi-level)	4.78×
Fairds-2	7.68×

1st-order is cheaper than bi-level; 2nd-order pays for the HVP.

When to use it

- **Structured bias, no group labels**, accuracy must be kept.
- A **closed-form** fairness signal — no inner loop — that preserves accuracy.
- **Not** a general tabular debiaser; use group-label methods when labels exist.

Contributions & takeaway

- A closed-form In-Run Data Shapley reweighter for **fair training** — down-weights biased examples, no group labels in the update.
- A **2nd-order cross-term with RMS rescaling** that targets the bias.
- A **residual ablation** proving the bias signal is a real direction, not smoothing.
- **Group fairness across 3 bias regimes**: worst-group ↑, majority–minority gap ↓ 40–47%; honest demographic boundary.

A closed-form data-value score can keep bias out of a model as it trains — within a clear regime.

Fairest among no-group-label methods; group-label / 2-stage methods remain stronger, and we say so.